

CSCI 1101 Introduction to Computer Science



NUMBER SYSTEMS

Digital Representation

1

Inside the computer...

Everything is discrete (digital), not continuous (analog) – *i.e.*, it must be represented using a discrete (number) value

Problem: How do we put the analog world “inside” the digital computer?

Solution: Devise ways to represent real-world, analog data in digital forms

But first... let's review some ideas about numbers.



Types of Numbers

2

Natural Numbers

Zero and any number obtained by repeatedly adding one to it.

Examples: 100, 0, 45645, 32

Negative Numbers

A value less than 0, with a – sign

Examples: -24, -1, -45645, -32

Integers

A natural number, a negative number

Examples: 249, 0, - 45645, - 32

Rational Numbers

An integer or the quotient of two integers

Examples: -249, -1, 0, $\frac{3}{7}$, $-\frac{2}{5}$

We need to be able to represent these different number types in the computer.

Natural Numbers

3

Puzzler:

What does the value 642 mean?

$600 + 40 + 2$?

Or is it

$384 + 32 + 2$?

Or maybe...

$1536 + 64 + 2$?

I know what you're thinking: "My professor is nuts. Everyone knows what that means." But DO WE?

Natural Numbers

4

Aha!

You probably assumed something: which **number system** we were using!

642 is $600 + 40 + 2$ in **BASE 10**, or in the *decimal* number system

The **base** of a number determines the **number of digits** and the **value of digit positions**

Positional Notation

5

Continuing with our example...

642 in **base 10 positional notation** is:

$$6 \times 10^2 = 6 \times 100 = 600$$

$$+ 4 \times 10^1 = 4 \times 10 = 40$$

$$+ 2 \times 10^0 = 2 \times 1 = 2 = 642 \text{ in base 10}$$

This number is in
base 10

The power indicates
the position of
the number

Positional Notation

6

What if 642 has the base of 13?

$$+ 6 \times 13^2 = 6 \times 169 = 1014$$

$$+ 4 \times 13^1 = 4 \times 13 = 52$$

$$+ 2 \times 13^0 = 2 \times 1 = 2$$

$$= 1068 \text{ in base 10}$$

642 in base 13 is equivalent to 1068
in base 10

The Binary System

7

The traditional **decimal** system is based on powers of **ten**.

The **binary** system is based on powers of **two**.

Binary

8

Decimal is **base 10** and has **10 digits**:

0,1,2,3,4,5,6,7,8,9

Binary is **base 2** and has **2 digits**:

0,1

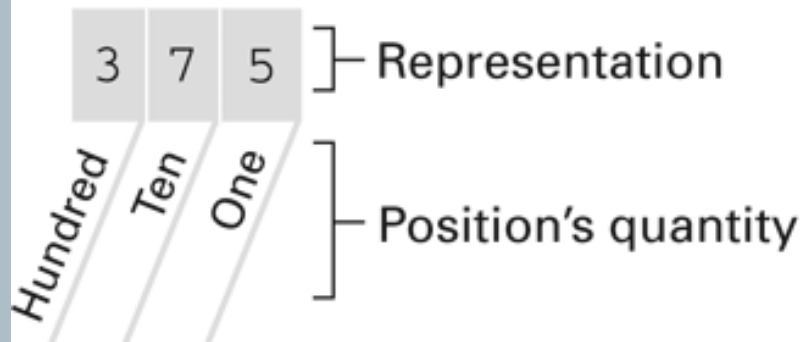
For a number to exist in a given base, it can only contain the digits in that base, which range from 0 up to (but not including) the base.

What bases can these numbers be in? 122, 198, 178, G1A4

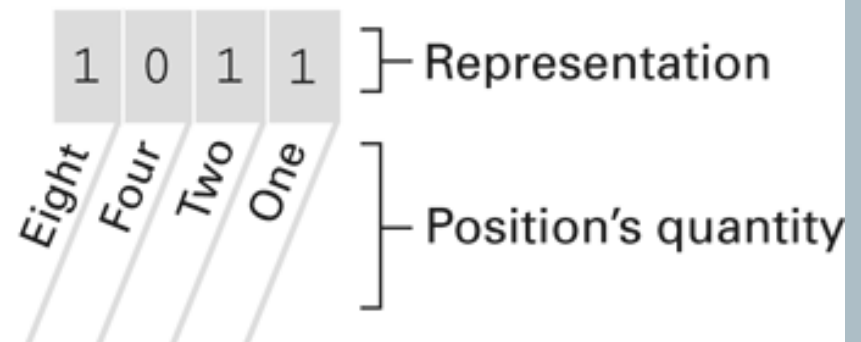
The decimal and binary systems

9

a. Base ten system



b. Base two system



Converting Binary to Decimal

10

What is the decimal equivalent of the binary number 1101110?

$$\begin{aligned} 1 \times 2^6 &= 1 \times 64 = 64 \\ + 1 \times 2^5 &= 1 \times 32 = 32 \\ + 0 \times 2^4 &= 0 \times 16 = 0 \\ + 1 \times 2^3 &= 1 \times 8 = 8 \\ + 1 \times 2^2 &= 1 \times 4 = 4 \\ + 1 \times 2^1 &= 1 \times 2 = 2 \\ + 0 \times 2^0 &= 0 \times 1 = 0 \\ &= 110 \text{ in base 10} \end{aligned}$$

Or, another way to look at it...

Decoding the binary representation 100101

11

Binary pattern [

1	0	0	1	0	1
				1	x one = 1
				0	x two = 0
			1		x four = 4
		0			x eight = 0
	0				x sixteen = 0
1					x thirty-two = <u>32</u>

Value
of bit

Position's
quantity

37 Total

Converting Decimal to Binary

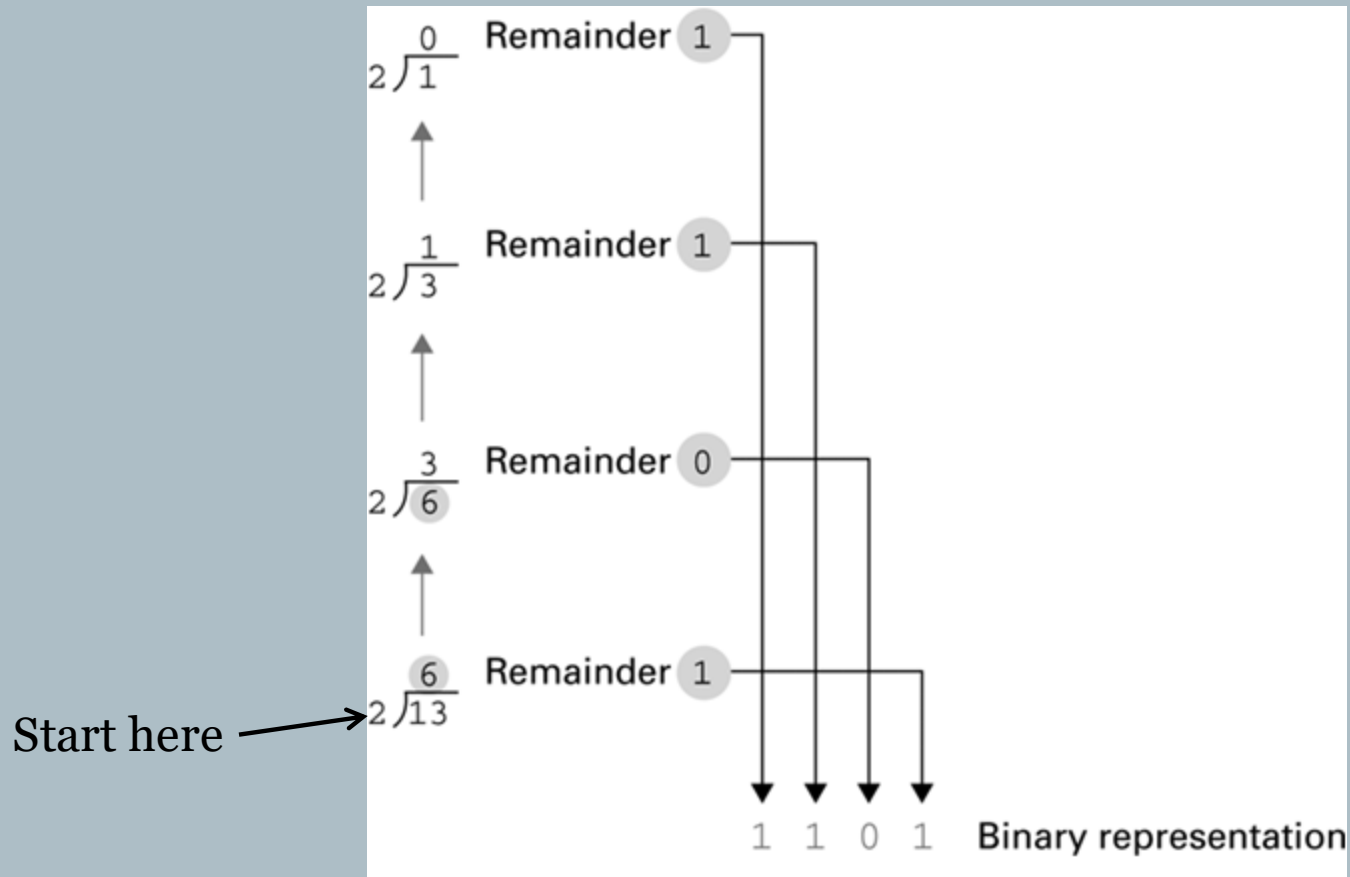
12

- Step 1.** Divide the value by two and record the remainder.
- Step 2.** As long as the quotient obtained is not zero, continue to divide the newest quotient by two and record the remainder.
- Step 3.** Now that a quotient of zero has been obtained, the binary representation of the original value consists of the remainders listed from right to left in the order they were recorded.

*This algorithm is generalizable. Just divide the decimal value by **ANY** new base to find the decimal equivalent in the new base!*

Applying the algorithm to obtain the binary representation of 13_{10}

13



Bases Higher than 10

14

How are digits in bases higher than 10 represented?

With distinct symbols for 10 and above.

Base 16 – hexadecimal – has 16 digits:

0,1,2,3,4,5,6,7,8,9,A,B,C,D,E, F

Why would we use such a strange base??

Hexadecimal and Binary are Related!

15

Bit pattern	Hexadecimal representation
0000	0
0001	1
0010	2
0011	3
0100	4
0101	5
0110	6
0111	7
1000	8
1001	9
1010	A
1011	B
1100	C
1101	D
1110	E
1111	F

Converting hexadecimal to binary is easy:

Just convert the hex digit to the corresponding binary digits (bits)

1 hex digit = 4 bits

Converting Binary to Hexadecimal

16

- Mark groups of *four* bits (starting at the right end)
- Convert each group of bits into hex digits

10101011 1010 1011
 A B

10101011 is AB in hexadecimal

Converting Hexadecimal to Decimal

17

What is the decimal equivalent of the hexadecimal number DEF?

$$\begin{aligned} D \times 16^2 &= 13 \times 256 = 3328 \\ + E \times 16^1 &= 14 \times 16 = 224 \\ + F \times 16^0 &= 15 \times 1 = 15 \\ &= 3567 \text{ in base 10} \end{aligned}$$

Remember, the digits in base 16 are
0,1,2,3,4,5,6,7,8,9,A,B,C,D,E,F

It's still just positional notation

Converting Decimal to Hexadecimal

18

What is 3567 (base 10) in base 16?

Use the algorithm from a few slides back (slide #12), dividing by 16 instead of 2

Try it!

Converting ANY Base to Decimal

19

To convert from ANY base to base 10:

Use the positional notation for the corresponding base.

Given a number in base B, to convert to base 10 use:

$$(d_n \dots d_2 d_1 d_0)_B = d_n \times B^n + \dots + d_2 \times B^2 + d_1 \times B^1 + d_0 \times B^0$$

$$= (\text{Value})_{10}$$

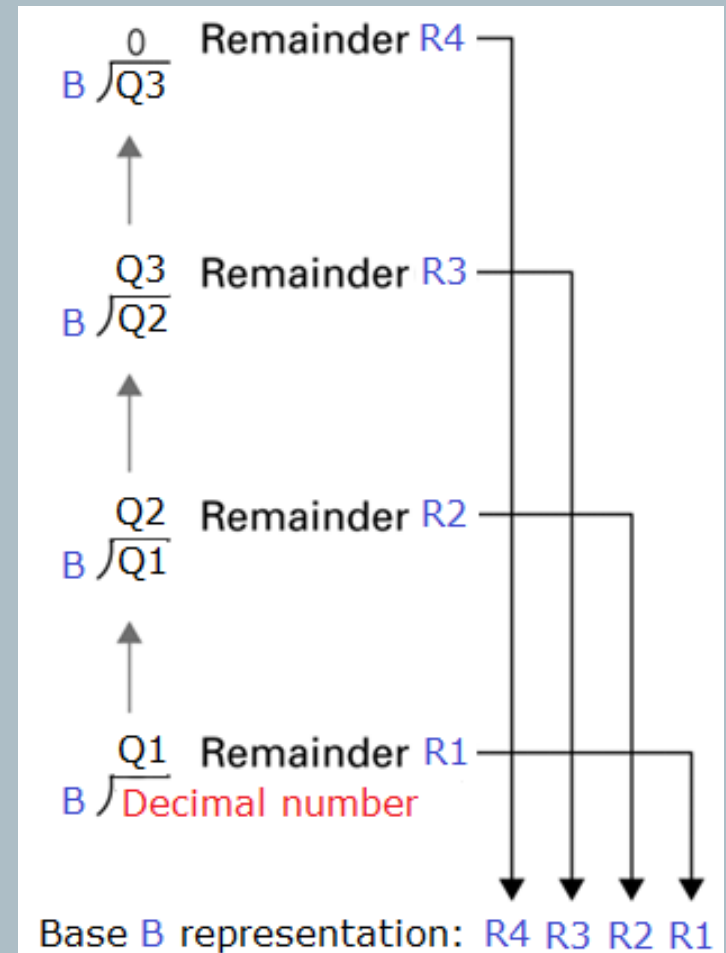
Converting Decimal to ANY Base

20

To convert from base **10** to **ANY** base:

Use the algorithm presented in slide 12 modified for the corresponding base.

Given a number in base **10**, to convert to base **B** use:



Why the Obsession with Binary and Hex?

21

Computers have storage units called **binary digits** or **bits**

The computer operates with voltage through what amounts to microscopic switches that are either “on” (1) or “off” (0)

Low Voltage = 0

High Voltage = 1 all bits have value 0 or 1